Sensors & Actuators: A. Physical 412 (2026) 118654

Contents lists available at ScienceDirect

Sensors and Actuators: A. Physical

journal homepage: www.journals.elsevier.com/sensors-and-actuators-a-physical

Thermal-plume and thermo-mechanical characteristics of Nitinol SMA actuators with bead-on-fiber coating

Raushan Kumar[a], Bal Krishan Mishra[b], Pradipta Kumar Panigrahi[a,*], Bijaya Bikram Samal[a], Ankur Dwivedi[a,c], Bishakh Bhattacharya[a,*]

[a] Department of Mechanical Engineering, IIT Kanpur, Kanpur, Uttar Pradesh, 208016, India
[b] National Wind Tunnel Facility, IIT Kanpur, Kanpur, Uttar Pradesh, 208016, India
[c] Department of Mechanical, Materials, and Aerospace Engineering, IIT Dharwad, Chikkamalligawad, Dharwad, Karnataka, 580011, India

HIGHLIGHTS

- The thermomechanical behavior of the SMA actuator is studied.
- Schlieren imaging, infrared thermography and a laser displacement sensor are used.
- The effect of SMA wire diameter and dip coating is systematically investigated.
- The actuator frequency increases with a reduction in SMA wire diameter and polymer coating.
- The normalized plume width and height show an exponential relationship, independent of wire diameter.

ARTICLE INFO

Keywords:
Shape memory alloy
Actuator
Coating
Schlieren-technique
Thermal plume
Heat transfer coefficient

ABSTRACT

Nitinol, a widely used shape memory alloy (SMA), offers a high power-to-weight ratio; however, its application is often limited by low actuation frequency due to thermal constraints. Effective thermal management, particularly through surface modification and flow control, is therefore critical. In this study, the buoyancy-driven natural convection generated by electrically heated Nitinol SMA wires of varying diameters is experimentally investigated using high-speed schlieren flow visualization and infrared (IR) thermography. The effect of polymer coating on SMA wire actuator performance is also investigated. The plume morphology is systematically analyzed by measuring the plume height and width. The results demonstrate that increasing wire diameter produces larger, wider, and more persistent thermal plumes. The normalized plume height and width show a universal correlation, which is independent of SMA wire diameter. The actuator frequency decreases with increases in SMA wire diameter. The heat transfer coefficient also decreases with an increase in the diameter of the SMA wire. Smaller-diameter SMA wires exhibit faster cooling and plume dissipation. The plume width is observed to be an order of magnitude higher than the diameter of the SMA wire. The plume width variation as a function of the wire diameter and actuation voltage from the present study provides guidelines for deciding the inter-wire spacing in multi-wire SMA actuators designed for higher force output. Furthermore, polymer dip coating of the SMA wire improves the actuation frequency and average mechanical energy output power compared with the uncoated wire, demonstrating that controlled surface modification is an effective strategy for enhancing the performance of SMA actuators.

1. Introduction

The shape memory alloy (SMA) is a class of smart materials well known for its high energy density, shape memory effect, and superelasticity. Owing to these unique characteristics, SMA has been widely employed in the development of linear and rotary actuators capable of generating high actuation forces relative to the recoverable strain. The compact size, lightweight, and high force-to-weight ratio make SMA actuators particularly attractive for space-constrained and weight-sensitive

* Corresponding authors.
Email addresses: raushank23@iitk.ac.in (R. Kumar), bkmishra@iitk.ac.in (B.K. Mishra), panig@iitk.ac.in (P.K. Panigrahi), bbsamal@iitk.ac.in (B.B. Samal), ankurdwi.vedant@iitdh.ac.in (A. Dwivedi), bishakh@iitk.ac.in (B. Bhattacharya).

https://doi.org/10.1016/j.sna.2026.118654
Received 10 August 2026; Received in revised form 29 September 2026; Accepted 14 October 2026
Available online 22 October 2026
0924-4247/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

applications, including aerospace, biomedical devices, robotics, and adaptive engineering systems [1–4]. Despite these advantages, the practical application of SMA actuators is often limited due to relatively low actuation frequency, which is primarily attributed to the slow cooling process following Joule heating. Electrical resistance heating is widely employed because of its simplicity, compactness, and ease of control [5,6]. Several approaches have been proposed to accelerate the heating process, including capacitor-based circuits [7] and high-current pulse excitation [8]. Nevertheless, the cooling stage remains the rate-limiting process governing the cyclic operating frequency of SMA actuators. Consequently, improving heat dissipation is one of the primary objectives in the development of high-speed and energy-efficient SMA actuation systems. To enhance cooling performance, numerous thermal management strategies have been explored, including natural air convection [9,10], liquid cooling [11,12], forced-air and evaporative cooling [13–15], spray cooling [16], geometric modifications [17], and magnetocaloric-assisted cooling [18].

Various coating materials and surface engineering approaches have emerged for tailoring the functional performance of SMA wires. Akash et al. [19] investigated copper-based SMA actuators deposited on Kapton polyimide substrates to improve actuation performance. Boccaccini et al. [20] fabricated PEEK and PEEK/Bioglass coatings on SMA wires using electrophoretic deposition (EPD) to improve biomedical performance. Marashi et al. [21] developed polydopamine-functionalized PPy/$BaTiO_3$ coatings on SMA to enhance corrosion resistance. Samal et al. [22] fabricated PMMA-coated SMA composites and demonstrated thermally induced bending actuation arising from the mismatch in the coefficients of thermal expansion between PMMA and SMA. Perez et al. [23] demonstrated PDMS/carbon nanopowder coatings as efficient photothermal layers, that enhance infrared light absorption and heat generation, enabling the SMA alloy to reach its phase transformation temperature at lower optical power and improving its light-driven actuation performance. Among the available surface modification techniques, dip coating offers a simple, scalable, and cost-effective approach for modifying SMA wire surfaces [24–27].

Several multi-wire actuator architectures have been proposed to enhance the actuation force. Britz et al. [28] developed parallel bundled SMA actuators comprising multiple wires operating within a compact constrained configuration to achieve higher force output. Barakat et al. [29] proposed an antagonistic multi-wire SMA actuator capable of high-frequency bidirectional rotary motion. Chaurasiya et al. [30] introduced a bio-inspired bipennate SMA actuator that reduced actuator weight while enhancing force amplification through hierarchical wire arrangement. These studies focused on actuator architecture, force generation, and system integration. Relatively little attention has been devoted to the transient thermal plume development around Joule-heated SMA wires under natural-convection conditions.

Natural convection around heated cylinders and wires has been extensively investigated because of its importance in thermal management, electronic cooling, and energy systems. The temperature difference between a heated surface and the surrounding fluid generates buoyancy forces leading to the formation of thermal plumes. Schlieren visualization has emerged as a powerful non-intrusive optical method for capturing refractive-index variations associated with temperature-induced density gradients. Fossa et al. [31] demonstrated the application of the schlieren technique for investigating natural-convection plume evolution around heated surfaces. The transport mechanisms governing buoyancy-driven convection, including plume development and thermal boundary layer growth, were comprehensively described by Gebhart et al. [32]. More recently, background-oriented schlieren techniques have enabled quantitative visualization of density gradients and thermal plumes in buoyancy-driven flows [33,34]. These techniques have been successfully applied to characterize plume evolution, wake dynamics, and natural-convection heat transfer around heated cylinders and plates [35–37].

It may be noted that SMA wire actuation involves both heating and cooling cycles in contrast to a heated wire configuration. Such buoyancy-driven flow characteristics directly influence the temporal evolution of the local convective heat transfer coefficient and wake dynamics, thereby limiting the cooling rate and cyclic operating frequency of SMA actuators. Understanding plume evolution is also important in multi-wire SMA actuator arrays, which are commonly employed to achieve higher actuation force or thrust. The above literature review shows that polymer-coated SMA actuators have previously been investigated to study mismatch-driven thermo-mechanical bending and photothermal actuation. However, the thermal plume development and thermomechanical behavior of the polymer coated and bare SMA actuators remain largely unexplored. In particular, the relationship between transient thermal plume evolution, buoyancy-driven heat transfer, and the thermo-mechanical performance of SMA actuators has not been investigated in the literature. The influence of polymer coatings on the transient thermal response and cooling behavior of Joule-heated SMA wires, the resulting thermal plume development and actuation frequency, has also received limited attention in the literature. Therefore, the present study provides a comprehensive experimental investigation of the coupled thermo-fluid and thermo-mechanical behavior of Joule-heated SMA wire actuators using a quantitative schlieren technique. In multi-wire configurations, the thermal plumes generated by neighboring wires may interact, altering the local temperature field and reducing convective cooling efficiency. Consequently, the spacing between adjacent SMA wires must also account for the plume expansion and thermal interference between neighboring wires. Therefore, quantitative characterization of transient thermal plume evolution provides a scientific basis for optimizing actuator spacing, minimizing thermal cross-talk, and improving the dynamic performance and energy efficiency of multi-wire SMA actuator arrays.

In the present study, SMA wires with diameters of 100, 150, 200, and 300 μm are investigated to elucidate the influence of wire diameter on thermal plume evolution, buoyancy-driven flow, and natural-convection heat transfer over an applied voltage range of 3–7 V. The thermal and mechanical responses of the actuators, including schlieren intensity distribution, temperature evolution, displacement, strain, heat transfer coefficient, and thermo-mechanical efficiency are reported. Furthermore, a dip-coated SMA wire exhibiting a bead-on-fiber morphology was compared with its uncoated counterpart to examine the influence of coating and surface morphology on transient plume development, wake dynamics, heat transfer and actuator frequency response. The correlations established between plume characteristics and actuator performance enhance the understanding of the coupled thermo-fluid mechanisms governing SMA actuation and are useful for the design of SMA actuators.

2. Methodology

The experimental setup used for SMA wire actuation, polymer dip coating, and schlieren flow visualization is described in this section. Subsequently, the data analysis procedures including image processing, plume parameter quantification, and the calculation procedure for the convective heat transfer coefficient are presented.

2.1. SMA wire actuation setup

Fig. 1 shows the schematic of the experimental setup used for SMA wire actuation, displacement measurement, and thermal imaging (the image of the actual experimental setup to measure temperature and displacement is shown in Figure S1(a) of the Supplementary Material). The experimental setup consists of an SMA wire, a fixed support, a movable slider integrated with a linear sliding bearing, a cylindrical guide rod, and a bias spring with a stiffness of 20 N/mm mounted on a wooden base. The SMA wire properties are provided in Table S1 of the Supplementary Material. One end of the SMA wire was attached to the fixed support, while the other end was connected to the movable slider through a pin. Upon Joule heating, the SMA wire contracted,

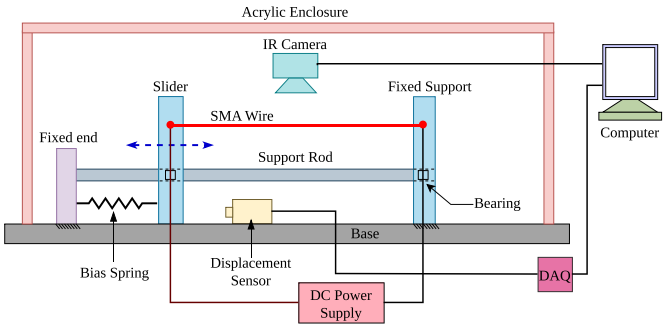


Fig. 1. Schematic of the experimental setup for the operation of the SMA wire actuator.

producing linear displacement of the slider, whereas the bias spring restored the wire to its initial position after the electrical supply was switched off. The SMA wires were electrically heated using a DC power supply operating in constant-voltage (CV) mode, with the applied voltage varying from 3 to 7 V. The power supply was controlled using a relay-based switching module interfaced with an Arduino-based microcontroller. For each SMA wire diameter, the heating time required for the SMA to reach the austenite finish temperature (A_f = 87.5 °C) was determined from the experimentally measured temperature response. The corresponding heating times for the 100, 150, 200, and 300 μm uncoated SMA wires were 1.01, 1.52, 1.86, and 2.30 s, respectively, while the heating time for the 150 μm coated SMA wire was 1.10 s. These experimentally determined heating times were used as the predefined switching durations for the Arduino-based relay system to switch off the power supply. Thus, the control scheme employed in the present study was a predefined time-based ON/OFF switching procedure rather than a closed-loop temperature-feedback controller.

The contraction in the SMA wire is measured by tracking the slider displacement using a laser displacement sensor (ILD1420-50, Micro-Epsilon, Germany). The sensor was aligned with the direction of slider motion and displacement data was continuously acquired using a computer through an RS422-to USB interface module (IF2001/USB, Micro-Epsilon, Germany). The surface temperature of the SMA wire was measured using a long-wave infrared (LW-IR) thermal camera (FLIR A655sc, FLIR Systems, USA) operating at a frame rate of 50 Hz. The camera was positioned at a distance of 24 cm from the SMA wire, and temperature measurements were obtained within a predefined region of interest (ROI). The IR images were recorded and analyzed using ResearchIR software provided by FLIR, USA. The IR measurements were carried out after spraying dry graphite lube onto the coated and uncoated SMA wires to ensure a high, and uniform surface emissivity [38,39]. Therefore, the temperature measurements remain unaffected by the emissivity of the polymer resin and NiTi material. Four different wire diameters (d) of 100, 150, 200 and 300 μm, each having an identical length (l) of 150 mm were used, and the actuation voltage is varied between 3 V and 7 V [40]. All the experiments were performed under controlled ambient laboratory conditions at the room temperature of 25 ± 1 °C. The temperature of the SMA wire at the beginning of each measurement was determined from the initial temperature recorded by the IR camera under the ambient experimental conditions.

2.2. Dip coating arrangement

The present study has been carried out using bare and in-house coated Flexinol SMA wire. The dip coating is carried out on the Flexinol SMA wire purchased from Dynalloy, Inc. under controlled process conditions. It may be noted that the coating produced in the present work is not a uniform cylindrical film. The objective was to generate a periodic beaded morphology on the 150 μm diameter SMA wire surface and examine its characteristics under varying withdrawal conditions. The dip-coating process generated a periodic bead-on-fiber morphology. Therefore, a single uniform coating-thickness value cannot adequately describe the resulting surface. The withdrawal velocity (v_w) was varied systematically to examine its influence on the resulting bead morphology and spacing along the SMA wire and the the coated actuator was prepared using coating condition corresponding to a withdrawal velocity of 2 mm s^{-1}. Under this condition, the measured bead pitch, axial bead length, and transverse bead width were approximately 800, 353, and 148 μm, respectively. These parameters describe the process-generated surface morphology.

Fig. 2 shows the schematic of the experimental arrangement to describe the dip coating procedure (the image of the actual experimental setup is shown in Figure S1(b) of the supplementary material). It consists of a vertically mounted SMA wire fixed on a motorized driver. The motion is restricted to the vertical (Z-axis) direction using a modified 3D printer mechanism, where the x and y axes remain stationary. The z-axis motion is controlled through G-code, allowing precise control of withdrawal velocity in the range of 1 mm s^{-1} to 3 mm s^{-1}. A high-speed camera (Chronos 2.1 by Kron Technologies) is employed to capture the bead formation dynamics during the coating process. A focused light source is used to enhance image contrast and improve visualization of the evolving bead structures. The lower end of the SMA wire is attached to a dead weight to maintain tension and ensure straight alignment during the experiment. The wire is immersed in a beaker (polymer tank) containing the polymer solution, which is shielded using an opaque covering to minimize any potential ambient light-induced polymerisation. *Elastic 600* resin (having density ~1.14 g cm^{-3} and viscosity ~450 mPa·s at 25°C), manufactured by MAGFORMS, was used as the coating material owing to its high elasticity, flexibility, toughness, and excellent fatigue resistance. Since the SMA is actuated by Joule heating, it is essential to ensure that the coating material remains thermally stable throughout the actuation process and does not undergo thermal degradation, which could adversely affect its structural integrity and functional performance during repeated actuation cycles. Therefore, the thermal stability of the resin was evaluated using thermogravimetric analysis (TGA). The corresponding TGA results are provided in the supplementary material; see Figure S2.

The TGA results revealed an initial weight loss of 5.8 wt.%, which is attributed to the evaporation of residual moisture and other volatile constituents. The resin subsequently underwent a major degradation stage with a mass loss of 87.8 wt.%, corresponding to the decomposition of the polymer network, while a residual mass of 6.4 wt.% remained at 800 °C, indicating the presence of thermally stable char or inorganic residues. The absence of significant mass loss prior to the main degradation stage confirms that the resin remains thermally stable within the temperature range employed for SMA actuation. After coating, the SMA wire was rinsed with acetone to remove excess polymer and subsequently cured using a rotating UV curing system (Anycubic UV Resin Wash & Cure Machine 2.0) for 20 min, ensuring uniform exposure and consistent polymer cross-linking. The geometric dimensions of the bead morphology, namely the pitch length (λ_p), bead length (b_l), and bead width (w_b), were measured from magnified microscopic images using ImageJ software. The measurements exhibited a standard deviation of approximately 50 μm. The corresponding measurements at different deposition velocities are presented in Table 1. Increasing the withdrawal velocity resulted in a reduction of the bead pitch, bead length, and bead width. This behavior is attributed to the thinner liquid film deposited at higher withdrawal velocities, which limits bead growth during the dip-coating process [41,42].

The thickness of the polymer coating is an important parameter governing the transient thermal response of the coated SMA actuator. The polymer layer introduces an additional radial thermal resistance between the SMA core and the surrounding environment, while simultaneously increasing the thermal capacitance of the actuator. Thus, the coating thickness represents a trade-off between modification of

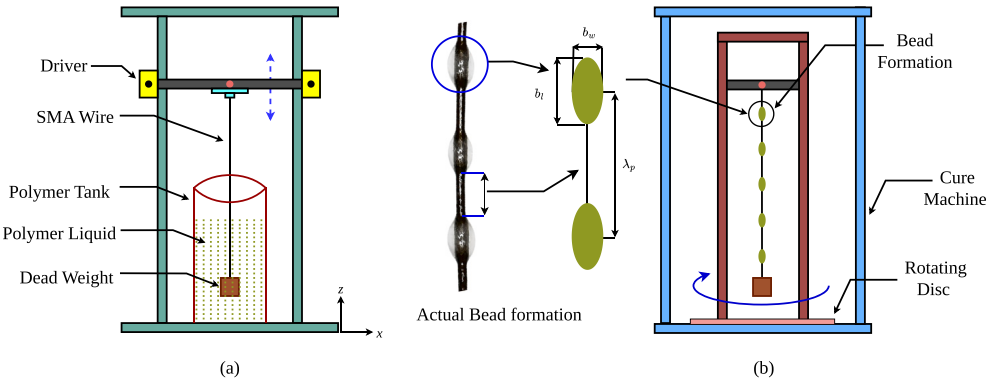


Fig. 2. Schematic of the coating setups: (a) dip-coating system and (b) UV-curing setup using the Anycubic UV resin Wash & Cure Machine. Here, λ_p denotes the pitch, l_b is the bead length (axial direction), and w_b is the bead width (transverse direction).

Table 1
Effect of withdrawal velocity (v_w) on bead geometry characteristics.

v_w (mm s^{-1})	λ_p (μm)	l_b (μm)	w_b (μm)
1	1200	586	187
2	800	353	148
3	760	246	126

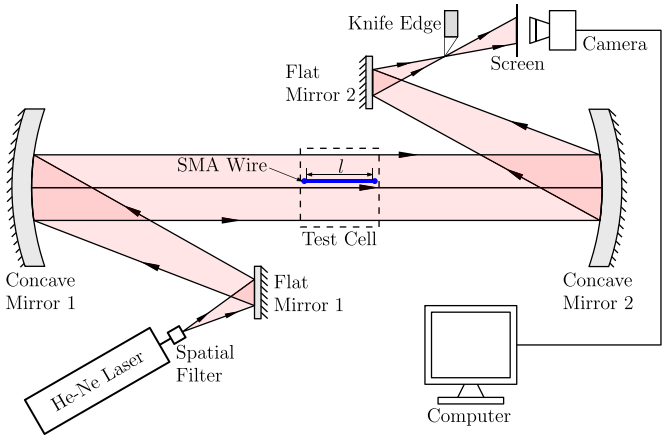


Fig. 3. Schematic of the experimental setup for the schlieren visualization.

the solid-fluid thermal boundary condition and additional thermal resistance. A very thin coating may produce only a limited modification of the surface thermal characteristics, whereas an excessively thick coating can increase the thermal time constant and slow the thermal cycling process. Therefore, coating thickness should be optimized according to the required actuation frequency and operating temperature rather than being increased indiscriminately. The polymer coating introduces an additional thermal resistance between the SMA core and the surrounding environment because of its relatively low thermal conductivity. However, the thermal response of the coated actuator is not determined by conduction through the coating alone. Heat generated within the SMA is conducted through the coating and subsequently transferred from the external coating surface to the surrounding fluid through convection and radiation. Consequently, the coating modifies the thermal boundary condition and the temperature distribution surrounding the actuator, which can influence the buoyancy-driven thermal plume and the associated cooling process.

2.3. Schlieren visualization setup

The thermal plume formed above the heated SMA wire due to natural convection driven by buoyancy forces and local fluid density changes is captured using a Z-type schlieren visualization setup. Fig. 3 shows the schematic of the schlieren visualization setup. The optical arrangement consists of a continuous-wave He–Ne laser (Melles-Griot 25-LHP-925-230), a spatial filter, a pair of concave mirrors, two flat mirrors, a knife-edge, a transparent screen, and a high-speed CMOS monochrome camera (MC1302, Mikrotron, Germany). The laser beam was first expanded using the spatial filter and reflected by a flat mirror onto the first concave mirror, which collimates the beam through the test section. The second concave mirror subsequently focused the beam onto the knife-edge positioned at its focal plane. More detailed information about the experimental setup for schlieren visualization can be found in Ref. [43]. The image acquisition was performed using a 64-bit frame grabber (X64-CL, Coreco Imaging, Canada) connected to a personal computer system. The sequence of schlieren images was captured at a frame rate of 400 frames per second with a spatial resolution of 336×360 pixels in video format. The video duration was set such that the natural convection phenomenon due to heating and cooling of the SMA wire is successfully captured in a single shot. To minimize disturbances from ambient airflow and ensure stable buoyancy-driven thermal plume formation, the entire setup was enclosed within a transparent acrylic enclosure. An acrylic chamber was used to provide a controlled and relatively quiescent environment for thermal-plume visualization and to minimize disturbances caused by external air currents, as shown in Fig. 1. It may be noted that the chamber is not airtight and the inside conditions remain the same as those of the ambient.

2.4. Image processing and plume characterization

The recorded high-speed schlieren videos were processed in MATLAB® to enhance the image quality and quantify the temporal characteristics of the plume during the heating and cooling of the SMA wire. Fig. 4(a) describes the procedure of image processing. The recorded video is first converted into a sequence of images. To eliminate the stationary optical features and non-uniform illumination effects, the background image was subtracted from the instantaneous image. The image just before the start of the actuation, which corresponds to the initial quiescent condition, was taken as the reference background image. For each subsequent frame, the absolute pixel-wise intensity difference was computed as given below:

$$\Delta I(x, y, t) = \left| I(x, y, t) - I_b(x, y) \right| \tag{1}$$

where $I(x, y, t)$, and $I_b(x, y)$ are the pixel intensity value of the instantaneous image at time t and base image, respectively at the location (x, y). The resulting intensity-difference field forms a two-dimensional matrix

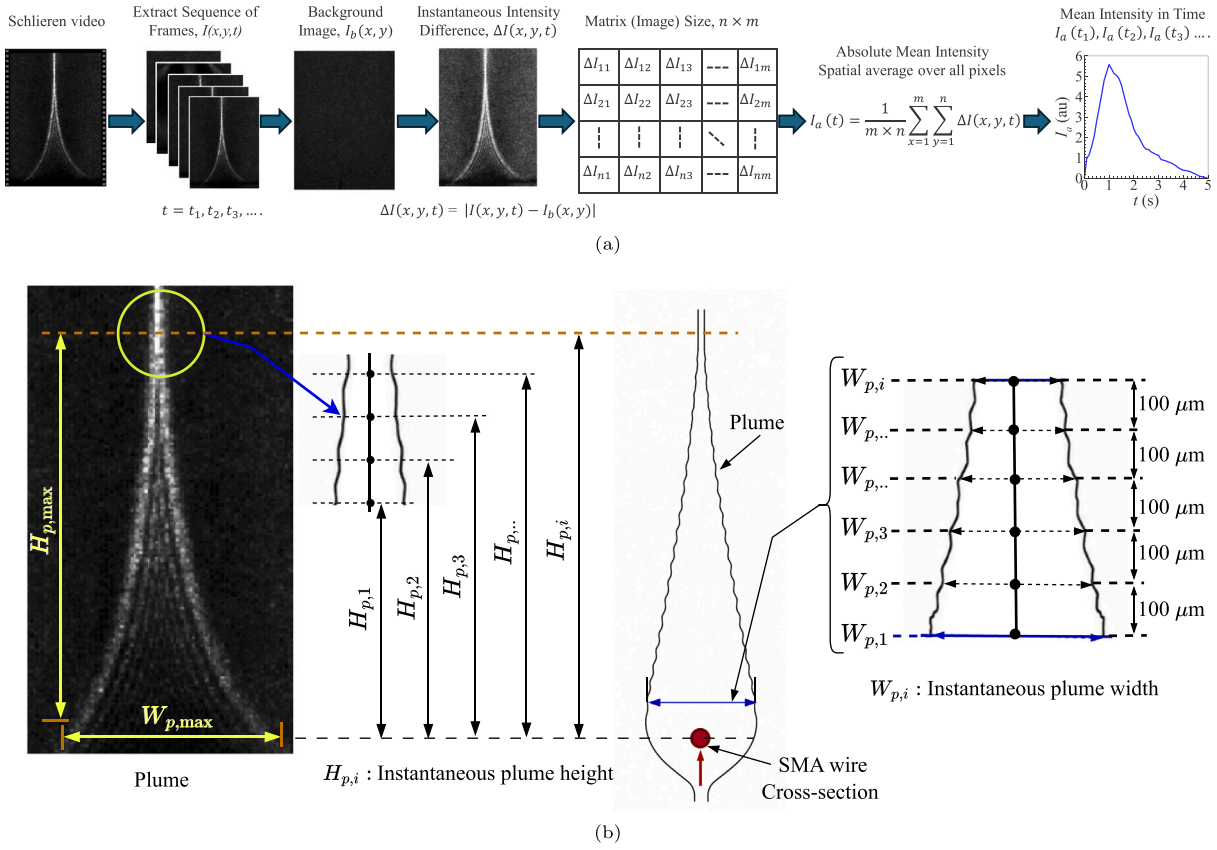


Fig. 4. Details of (a) Schlieren image-processing, and (b) Measurement of plume size and near field plume profile.

of size $(n \times m)$, where m and n denote the number of pixels in x and y-axis directions, respectively. To obtain a single representative average intensity of each instantaneous image, the intensity-difference field was spatially averaged over all pixels as mentioned below:

$$I_a(t) = \frac{1}{m \times n} \sum_{x=1}^{m} \sum_{y=1}^{n} \Delta I(x,y,t) \tag{2}$$

The heated SMA wire generates temperature gradients in the surrounding air, which in turn produces density gradients. These density gradients induce spatial variations in the refractive index, causing light rays to deviate from their original paths. The resulting light deflections appear as brightness and contrast variations in the recorded schlieren images. Consequently, regions of stronger thermal and density gradients produce larger intensity differences relative to the background. Therefore, the absolute mean intensity of the schlieren image at a given instant $I_a(t)$, represents the strength of the thermal plume at the given time instant. Thus, the temporal evolution of the plume can be understood by plotting I_a as a function of time (see Fig. 4(a)).

The plume generated around the SMA wire during the heating and cooling processes is quantified from schlieren images by measuring the plume width and plume height. Fig. 4(b) illustrates the calculation procedure details for the plume size and geometry. A rectangular region of interest (ROI) containing the thermal plume was selected to encompass the entire visible plume around the SMA wire. The same ROI dimensions were maintained for the entire image sequence corresponding to a given test condition. The selected ROI was subsequently processed using a Gaussian filter to enhance the visualization of the thermal plume and facilitate the identification of the plume boundaries. The vertical SMA wire axis (same as the plume central axis) and the horizontal SMA wire axis are considered the reference for the plume width and plume height measurements, respectively. Equally spaced measurement sections were selected at intervals of 100 μm along the plume-development region, starting about 1 mm above the wire surface. The plume width is measured as the distance between the left and right plume boundaries identified from the processed schlieren images. The instantaneous plume width, $W_{p,i}$ and plume height, $H_{p,i}$ measurements are temporally averaged over three consecutive schlieren frames as: $W_{p,i} = \frac{1}{3}\sum_{t=1}^{3} W_{p,i}$ and $H_{p,i} = \frac{1}{3}\sum_{t=1}^{3} H_{p,i}$ to reduce the measurement uncertainty arising from the schlieren image fluctuations. Since the plume width varies along the y-axis because of the lateral spreading of the plume, the average width obtained from the five measurement sections starting from the base is reported as the maximum plume width, $W_{p,max}$. The maximum plume height, $H_{p,max}$ is taken as the vertical distance between the SMA wire horizontal axis and the plume boundary merging point, (i.e., the location at which the left and right plume boundaries converge and the plume width becomes constant, see Fig. 4(b)).

2.5. Calculation of heat transfer coefficient

The Nusselt number, $Nu = hd/k$, where d is the characteristic length, h is the heat transfer coefficient, and k is the thermal conductivity of the fluid, is a function of $GrPr$, where $Gr = g\beta(Ts - T_\infty)d^3/\nu^2$ is the Grashof number and $Pr = \nu/\alpha$ is the Prandtl number. Here g is the acceleration due to gravity; $\beta = 1/T_f$ is the thermal expansion coefficient of the fluid; T_s is the material surface temperature; T_∞ is the ambient fluid (i.e., air) temperature; $T_f = (T_s + T_\infty)/2$ is the film temperature; α is the thermal diffusivity of the fluid; and ν is the kinematic viscosity of the fluid. The product of Grashof number and Prandtl number is called the Rayleigh number, Ra. The values of the non-dimensional numbers are calculated using the fluid properties at the film temperature. [44] obtained the Nu correlation for natural convection around a horizontal cylinder as given

below:

$$Nu = \left[0.60 + \frac{0.387 Ra^{1/6}}{\left(1 + \left(\frac{0.559}{Pr}\right)^{9/16}\right)^{8/27}} \right]^2 \quad (3)$$

Eq. (3) is valid for $0 < Ra \leq 10^{12}$. This empirical relationship is used to estimate the heat transfer coefficient from the SMA wire.

In the present study, the experimental values of the heat transfer coefficient are also estimated by comparing the analytical solution of Newton's law of cooling based on the lumped-capacitance model (Eq. 4) with the exponential curve fitting (Eq. 5) applied to the experimental values of instantaneous wire surface temperature, $T(t)$. The solution of Newton's law of cooling for the instantaneous temperature is given as:

$$T(t) = T_\infty + \left(T_0 - T_\infty\right) \exp\left(-\frac{hA_s}{\rho c_p V} t\right) \quad (4)$$

where, t is the time; T_0 is the initial wire surface temperature; and ρ, c_p, A_s, and V are the density, specific heat capacity, surface area, and volume of the material, respectively. The curve fitting equation is given as:

$$T(t) = T_\infty + \left(T_0 - T_\infty\right) e^{-\tau t} \quad (5)$$

Here, τ is the time constant which is obtained by fitting an exponential curve to the experimental data. From the comparison of Eqs. (4) and (5), we can deduce that the heat transfer coefficient can be expressed as:

$$\tau = \frac{hA_s}{\rho c_p V} \Rightarrow h = \frac{\tau \rho c_p V}{A_s} \quad (6)$$

For a cylindrical wire, $V = \pi d^2 L/4$, and $A_s = \pi d L$, where d is the diameter of wire, and L is the length of wire. Substituting the value of V/A_s into Eq. (6), we get the final expression for the experimental estimation of h as given below,

$$h = \frac{\tau \rho c_p d}{4} \quad (7)$$

The thermal behavior of the SMA wire is characterized using the Rayleigh number and the Nusselt number. Later, the average heat transfer coefficients for the SMA wire cooling period, obtained from the correlation of [44] (Eq. 3) and the lumped-capacitance analysis (Eq. 7) are compared.

3. Results and discussion

The heat transfer and actuator characteristics as a function of heating rate, wire diameter, and coating on the SMA wire are discussed in this section.

3.1. Heating rate and actuation behavior

The power supply of the SMA actuator was operated in constant voltage mode for a prescribed duration of 2 s while keeping the maximum current limit set to 2 A. However, the actual current passing through the NiTi wire varies during heating as its electrical resistance changes with temperature due to phase transformation. For the 0.15 mm SMA wire, the measured current ranged from 0.10–0.12 A, 0.15–0.17 A, 0.22–0.24 A, 0.28–0.30 A, and 0.33–0.35 A at applied voltages of 3, 4, 5, 6, and 7 V, respectively. This heating duration was adopted throughout the experiments to provide a consistent basis for comparing the thermal response of SMA wires with different diameters under identical electrical loading conditions while avoiding excessive overheating at higher applied voltages [7,45].

Fig. 5 shows the effect of applied voltage on the thermal and actuation characteristics of SMA wires for the four different wire diameters. At 3 V, it can be observed that none of the investigated SMA wires reached the austenite finish temperature (A_f = 87.5 °C) within the prescribed duration of two seconds. Consequently, complete phase transformation was not achieved under these low-voltage operating conditions. The applied voltage was subsequently increased, resulting in a corresponding increase in the SMA temperature. The heating rate increases with increasing activation voltage owing to enhanced Joule heating, enabling the wires to reach the austenite finish temperature in less than 2 s. At an applied voltage of 7 V, all the wire diameters reached the austenite finish temperature ($\Delta T = 65$°C) within the prescribed duration i.e., 2 s. The time required to attain the transformation temperature increases with wire diameter because of the higher thermal mass associated with thicker wires and lower resistance (see Figure S3 in the supplementary material, as shown in Fig. 5(b)).

Fig. 5(c) shows the maximum axial contraction of the SMA wires obtained from the displacement measurements as a function of the applied voltage. These results indicate that the maximum recoverable strain of approximately 4% is achieved at an applied voltage of 7 V for all wire diameters which corresponds to a complete austenite transformation. At lower voltages, particularly at 3 V, the thinner SMA wires exhibited greater initial contraction than the larger-diameter wires because of lower thermal inertia and faster heating response. This observation demonstrates that smaller-diameter SMA wires are capable of achieving measurable actuation at lower voltage levels, whereas larger-diameter wires require higher energy input to initiate significant phase transformation and contraction.

Fig. 5(d) shows the variation in the heating rate, calculated as $(\Delta T/t_h)$, where $\Delta T = T_{max} - T_{atm}$ and Δt denotes the heating time required for the SMA wire to reach its maximum temperature. As the applied voltage increases, the Joule heating input increases, resulting in a larger temperature rise while simultaneously reducing the heating time. Consequently, the heating rate increases with increasing applied voltage for all SMA wire diameters. The 100 μm diameter wire exhibits the highest heating rate, increasing from approximately 13 °C/s at 3 V to about 64 °C/s at 7 V. In contrast, the 300 μm wire shows the lowest heating rate, increasing from approximately 4 °C/s to 31 °C/s over the same voltage range. This behavior is attributed to the lower thermal mass and higher electrical resistance (refer to Figure S3 in the supplementary material) of the thinner SMA wires, which enable a rapid temperature rise for a given electrical input. As the wire diameter increases, the thermal mass increases and the electrical resistance decreases, resulting in slower heating and lower heating rates.

3.2. Schlieren visualization

Fig. 6 shows the plume development at different stages of heating and cooling as a function of SMA wire diameter. During the initial heating stage ($t = 0.25t_h$), a weak thermal plume forms above the wire as the surrounding air begins to rise due to buoyancy. As heating progresses ($t = 0.5t_h$ and $0.75t_h$), the plume becomes more distinct, exhibiting a vertically elongated structure with gradual lateral spreading. At the end of the heating stage (t_h), the plume attains its maximum height and intensity. During cooling ($0.25t_c$–$0.75t_c$), the plume gradually weakens, contracts, and eventually disappears as the temperature difference between the SMA wire and the ambient air decreases. Throughout the heating and cooling cycle, the plume remains nearly symmetric about the wire centerline, indicating stable laminar buoyancy-driven flow.

There is a clear dependence of plume morphology on wire diameter. The 100 μm wire generates a relatively short and narrow plume that dissipates rapidly during cooling, whereas the 200 μm and 300 μm diameter wires produce longer, wider, and more persistent thermal plumes owing to greater thermal inertia. The 150 μm coated wire exhibits a comparatively weaker and less extensive plume than the uncoated wire, indicating that the coating modifies the heat transfer from the wire surface and consequently alters the surrounding buoyancy-driven flow field.

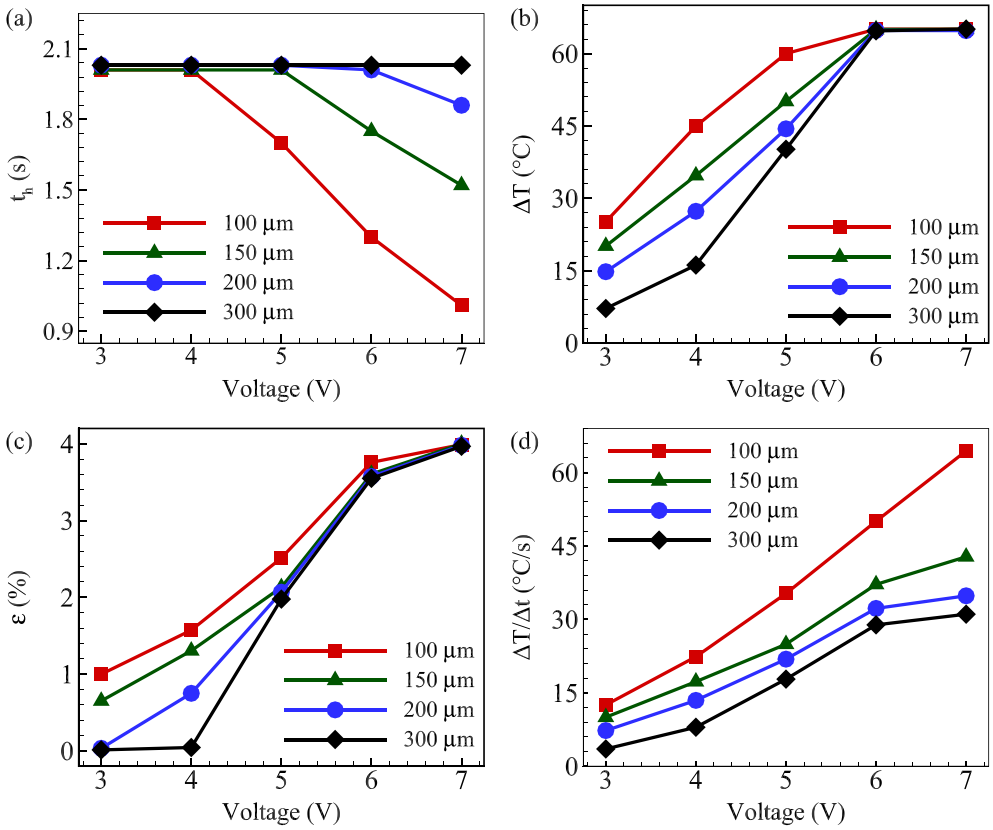


Fig. 5. Effect of applied voltage on the thermo-mechanical behavior of SMA wires for different diameters: (a) voltage activation time (t_h), (b) steady-state temperature rise (ΔT), (c) maximum axial contraction (ε), and (d) heating rate ($\Delta T/\Delta t$).

3.3. Plume characterization

The temporal evolution of the schlieren intensity for different SMA wire diameters is shown in Fig. 7(a). For all investigated wire diameters, the intensity difference increases during the heating stage and reaches a maximum near the peak wire temperature. The maximum intensity difference decreases with increasing wire diameter, reaching approximately 5.5, 4.7, 4.0, and 3.1 a.u. for the 100, 150, 200, and 300 µm wires, respectively. Furthermore, the time required to reach peak intensity increases with an increase in wire diameter, and the subsequent decay becomes more gradual with increasing wire diameter, indicating that larger-diameter wires retain heat for a longer duration and sustain the thermal plume over an extended period.

Fig. 7(b) shows the normalized plume height versus width of SMA wires at a maximum wire temperature of approximately 90°C. The normalized data for all wire diameters collapse onto a single generalized curve, demonstrating a universal relationship between plume height and plume width. The results indicate that the plume height decreases exponentially with increasing plume width and can be expressed as $H^* = Ae^{-BW^*}$, where W^* is the normalized plume width, H^* is the normalized plume height, and A and B are fitting constants. The parameter A remains close to unity $A = 1.11 \pm 0.03$, while the exponential decay constant is $B = 2.89 \pm 0.14$. The consistency of the fitting parameters across all wire diameters suggests that the plume geometry follows a common exponential scaling law, independent of wire diameter.

3.4. Thermo-mechanical characteristics

The preceding discussion established the evolution of the thermal plume generated during the heating and cooling of SMA wires. In this section, we investigate the relationships among the plume intensity difference (ΔI_a), strain (ϵ), and temperature difference (ΔT). Fig. 8 shows that these three parameters exhibit a distinct peak corresponding to the austenite finish temperature ($T_{A,f}$) of the SMA, indicating a strong correlation between the thermo-mechanical phase transformation and the evolution of the thermal plume. The maximum intensity difference, peak temperature difference, and maximum recoverable strain occur almost simultaneously at approximately 2.30 s, marking the end of the heating stage and the completion of the martensite-to-austenite phase transformation. The cooling time is about 11 s compared to the 2.3 s heating time. The similar nature of these characteristic parameters demonstrates that the evolution of the thermal plume is intrinsically coupled with the thermo-mechanical response of the SMA wire. This strong coupling suggests that the thermal plume characteristics obtained from schlieren imaging can serve as a non-contact diagnostic tool for monitoring the phase transformation and actuation behavior of SMA actuators in real time.

3.5. Heat transfer characteristics

The heat transfer characteristics are analyzed using the Rayleigh number (Ra), Nusselt number (Nu), and the convective heat transfer coefficient (h) to characterize the thermo-fluid behavior surrounding the SMA wires under natural-convection conditions. The numerical values of these quantities are calculated using the temperature values obtained from IR thermography. The calculation procedure for the convective heat transfer coefficient is described in Section 2.5. The Rayleigh number characterizes the strength of buoyancy-induced flow, while the Nusselt number represents the enhancement of convective heat transfer relative to conduction. The convective heat transfer coefficient obtained from the lumped-capacitance analysis is further compared with the analytical values calculated from the [44] correlation for natural convection around SMA wires. This comparison was performed to assess the applicability of the lumped-capacitance model during the cooling process.

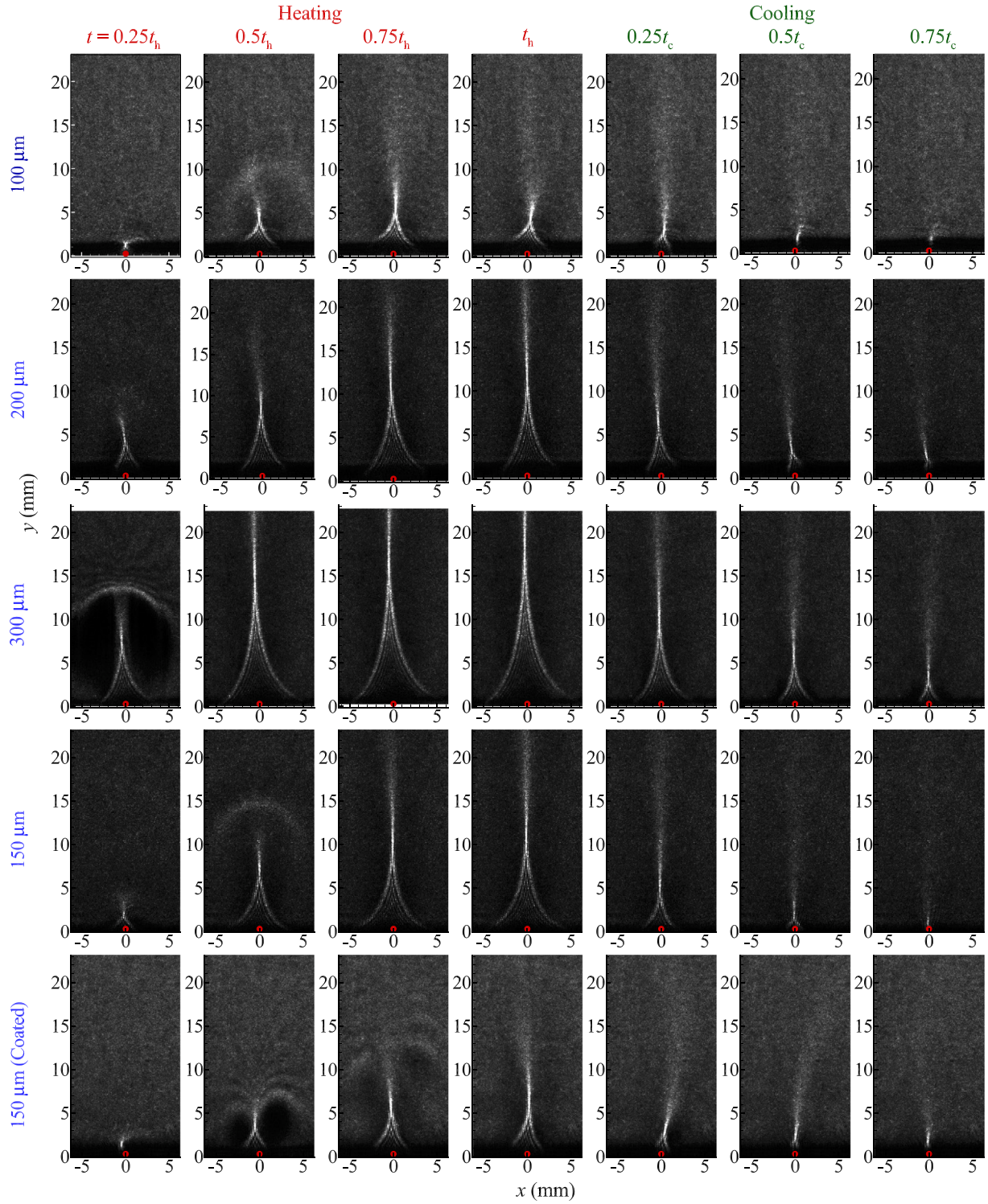


Fig. 6. Instantaneous schlieren visualization of the plume generated by the SMA wire for different diameters of the SMA wire, i.e, 100 µm, 150 µm, 200 µm, 300 µm, and 150 µm polymer-coated wire at an applied voltage of 7 V. For the heating cycle, schlieren images are presented at 0.25, 0.5, 0.75, and 1 times the total activation duration, and for the cooling cycle, images are shown at 0.25, 0.5 and 0.75 times the total cooling duration.

Fig. 9(a) shows the temporal variation of the Rayleigh number for different SMA wire diameters. It can be observed that the Rayleigh number initially increases rapidly during the heating stage, reaches a peak value, and subsequently decreases during cooling. The peak Rayleigh number is equal to 0.11, 0.03, 0.014, and 0.004 for SMA wire diameters of 300 µm, 200 µm, 150 µm and 100 µm, respectively. The maximum Rayleigh numbers obtained for all investigated wire diameters are well below the critical value for the onset of flow instability. Therefore, the buoyancy-induced flow remains within the laminar natural convection regime throughout the heating and cooling cycle. This indicates that heat transfer around the SMA wires is in the laminar regime. The higher peak value observed for the 300 µm wire is associated with its larger

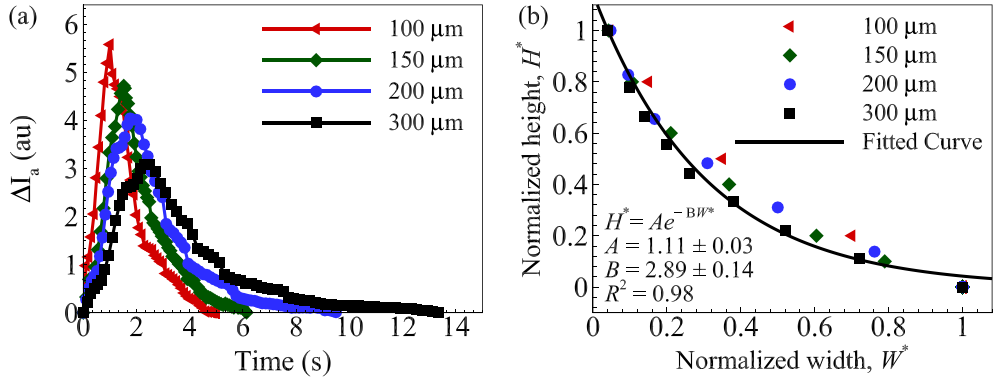


Fig. 7. (a) Temporal variation of the averaged intensity (I_a), and (b) the nondimensional plume height ($H^* = H_{p,i}/H_{p,\max}$) versus the nondimensional plume width ($W^* = W_{p,i}/W_{p,\max}$) for four different SMA wire diameters (µm): 100, 150, 200, and 300 with an actuation voltage of 7 V.

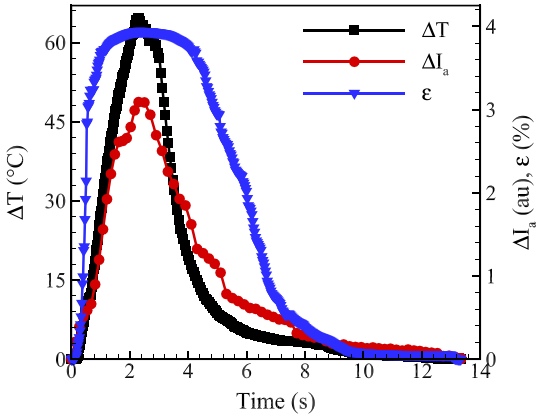


Fig. 8. Temporal variations of temperature difference, average intensity difference, and strain for the 300 µm diameter SMA wire during heating at 7 V and subsequent cooling under natural convection.

plume dimensions and prolonged thermal response compared with the smaller-diameter wires. A noticeable shift in the peak location is also observed with increasing wire diameter. The smaller-diameter wires attain the maximum Rayleigh number at earlier times. Following the peak, there is a gradual decay in the Rayleigh number during cooling. The rate of decay is a function of wire diameter. The 100 µm wire exhibits the most rapid decrease.

Fig. 9(b) illustrates the temporal evolution of the Nusselt number (Nu) for SMA wires of different diameters, calculated using Eq. (3). The Nusselt number initially increases during the heating stage as the wire temperature rises, reflecting the progressive enhancement of buoyancy-driven natural convection and the associated development of the thermal plume. As the temperature difference between the SMA wire and the surrounding air reaches its maximum, the convective heat transfer rate is enhanced, resulting in the peak Nusselt number. During the subsequent cooling stage, the Nusselt number gradually decreases owing to the reduction in the temperature gradient, which weakens the buoyancy forces and consequently reduces the convective heat transfer rate. The maximum Nusselt number increases with wire diameter, reaching values of 0.52, 0.56, 0.60, and 0.67 for the 100, 150, 200, and 300 µm SMA wires, respectively. Since the Nusselt number remains below unity for all investigated wire diameters, heat transfer is predominantly conduction-dominated, with natural convection providing only a modest enhancement. Nevertheless, the increase in the Nusselt number with wire diameter indicates a progressively stronger contribution of buoyancy-driven convection to the overall heat transfer. Furthermore, the peak Nusselt number occurs progressively later with increasing wire diameter, indicating a slower thermal response due to the greater thermal inertia of thicker SMA wires.

Fig. 9(c) shows a comparison of the transient temperature evolution from the lumped-capacitance method using Eq. (5) with that of the experiment for a 100 µm-diameter SMA wire. The close match between the two indicates the validity of the lumped-capacitance model during the cooling of the SMA wire. Fig. 9(d) compares the average convective heat transfer coefficients obtained from the Churchill–Chu natural convection correlation and the lumped-capacitance model calculated using Eq. (7). For the Churchill–Chu approach, the instantaneous convective heat transfer coefficient was evaluated from the transient Rayleigh number using the instantaneous temperature difference between the SMA wire and the ambient air. The resulting time-dependent heat transfer coefficient was subsequently averaged over the entire cooling period using Simpson's 1/3 numerical integration rule. Both methods consistently predict an increase in the heat transfer coefficient with decreasing wire diameter, confirming the enhanced cooling performance of thinner SMA wires under natural convection. Although both approaches exhibit the same diameter-dependent trend, the effective heat transfer coefficients obtained from the lumped-capacitance model are consistently higher than those predicted by the Churchill–Chu correlation. The percentage differences between the two estimates are approximately 27.6%, 27.2%, 23.7%, and 20.9% for the 100, 150, 200, and 300 µm diameter SMA wires, respectively. The higher value of the heat transfer coefficient in the lumped capacitance approach may be attributed to the dynamic movement of the SMA wire during actuation. Despite these methodological differences, both approaches demonstrate a consistent wire-diameter-dependent trend of the SMA wire.

3.6. Effect of coating

A polymer-coated 150 µm diameter SMA wire is used to investigate the effect of polymer coating on the thermal plume and thermo-mechanical characteristics. The polymer coating modifies the heat transfer between the SMA wire and the surrounding air through the formation of a bead-like surface morphology, which enhances natural convection and accelerates the cooling process. Consequently, the heating and cooling times, plume geometry (width and length), and the corresponding thermo-mechanical response of the SMA wire are also affected. The influence of the polymer coating on these characteristics is discussed in the following sections.

Fig. 10(a) compares the heating (actuation) time of coated and uncoated 150 µm diameter SMA wires as a function of applied voltage. As the applied voltage increases, the electrical power input increases, resulting in a higher Joule heating rate and a corresponding reduction in the time required for the SMA wire to reach the austenite finish temperature. Consequently, the actuation time decreases for both coated

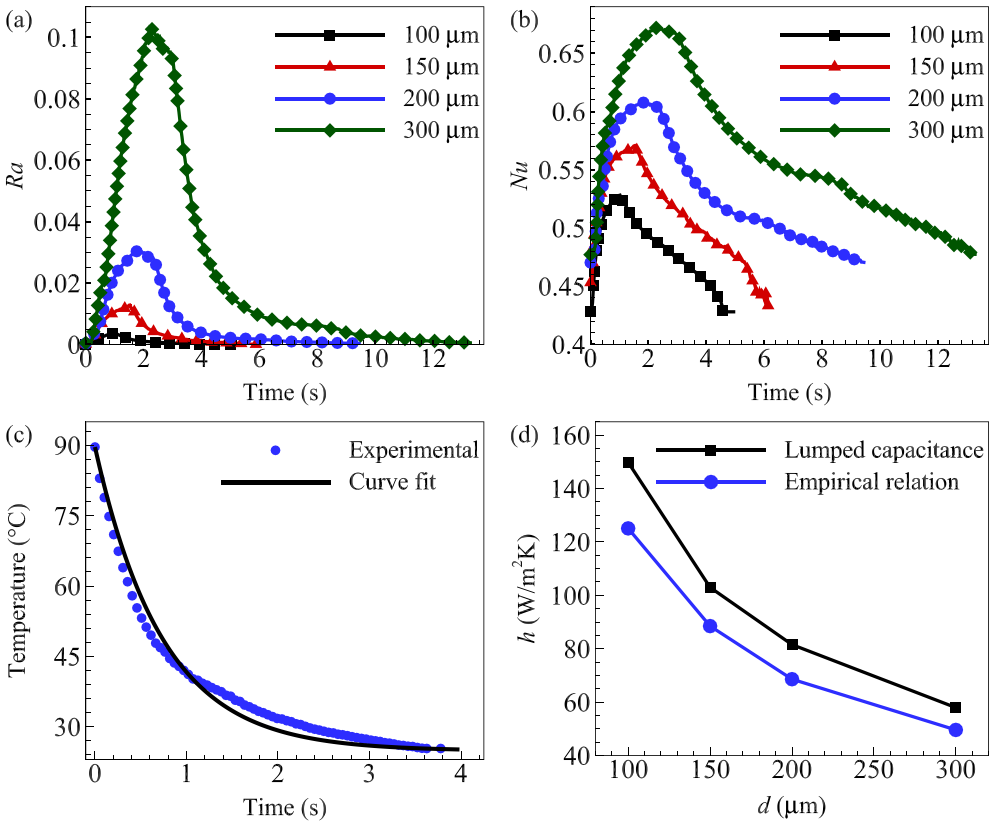


Fig. 9. Heat transfer non-dimensional parameters and heat transfer coefficient of SMA wires as a function of SMA wire diameter: (a) Rayleigh number, (b) Nusselt number, (c) comparison between the experimental temperature variation and the lumped capacitance model for a 100 µm diameter SMA wire, and (d) comparison of the heat transfer coefficients obtained from the lumped capacitance model and the Churchill–Chu correlation.

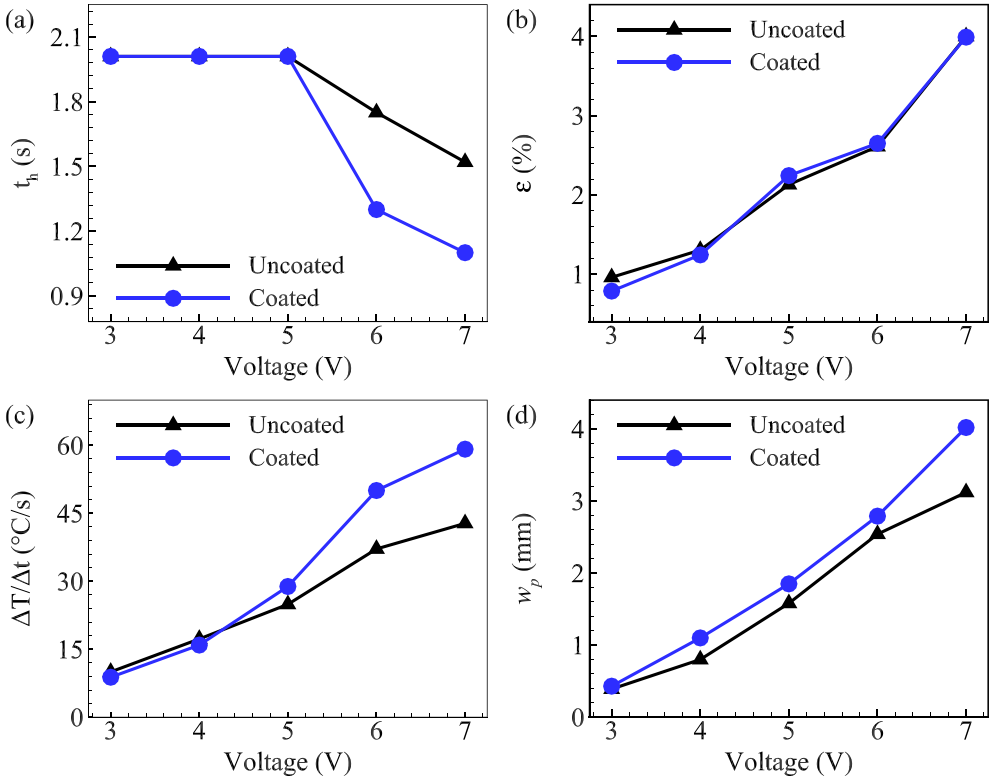


Fig. 10. Comparison for thermo-mechanical characteristics of uncoated and coated 150 µm diameter SMA wires as a function of applied voltage: (a) voltage activation time (t_h), (b) maximum axial contraction (ε), (c) heating rate ($\Delta T/\Delta t$), and (d) thermal plume width (w_p).

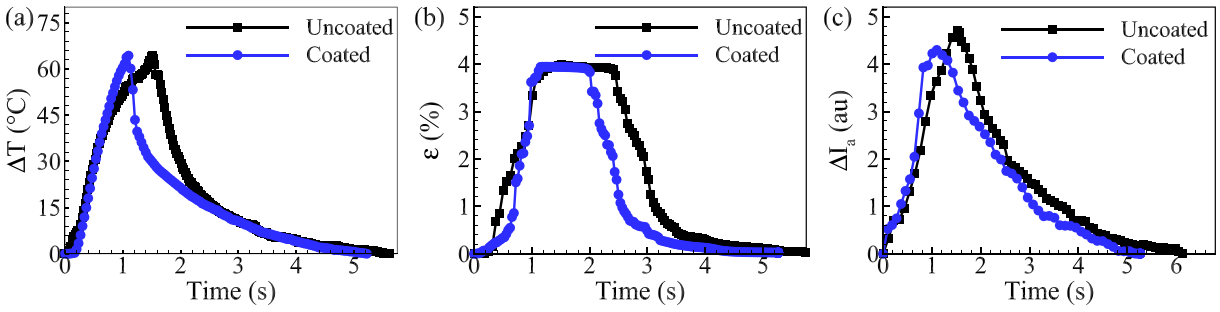


Fig. 11. Comparison of the temporal evolution of (a) temperature change (ΔT), (b) axial contraction (ε), and (c) average intensity difference (ΔI_a) for uncoated and coated 150 μm diameter SMA wires at an applied voltage of 7 V and during subsequent cooling under natural convection.

and uncoated wires with increasing voltage. The heating time of the coated SMA wire is higher than that of the uncoated SMA wire due to additional thermal resistance and lower heat transfer from the coated SMA wire. The bead-like surface morphology of the polymer coating enhances heat transfer due to three dimensionality introduced by the bead. Consequently, the coated SMA wire exhibits a consistently shorter actuation time than the uncoated wire. At an applied voltage of 7 V, the actuation time decreases from 1.52 s for the uncoated wire to 1.10 s for the coated wire.

Fig. 10(b) shows the variation of the recoverable strain (ε) with applied voltage for the uncoated and coated 150 μm SMA wires. For both wires, the recoverable strain increases progressively with increasing voltage as the higher Joule heating raises the wire temperature, promoting a greater martensite-to-austenite phase transformation and consequently a larger recovery strain. The coated and uncoated wires exhibit nearly identical strain values over the entire voltage range, reaching a maximum recoverable strain of approximately 4% at 7 V. This indicates that the polymer coating does not affect the transformation strain or the maximum mechanical deformation of the SMA wire. Instead, the primary effect of the coating is to improve the thermal response by accelerating the heating and cooling processes, resulting in a higher actuation frequency while maintaining nearly the same recoverable strain.

Fig. 10(c) compares the heating rate, calculated as ($\Delta T/t_h$), of the uncoated and coated 150 μm SMA wires under different applied voltages. The heating rate increases with increasing voltage for both wires due to the higher Joule heating input. The coated SMA wire exhibits a consistently higher heating rate, increasing from approximately 9 °C/s at 3 V to about 59 °C/s at 7 V. In contrast, the uncoated wire shows a heating rate change from 10 °C/s to 43 °C/s. The enhanced heating rate is attributed to the polymer coating, which reduces the heat loss from the SMA wire due to the lower thermal conductivity of the polymer coating, leading to an increase in conduction resistance.

Fig. 10(d) shows the variation in the thermal plume width (w_p) for the uncoated and coated 150 μm SMA wires. The plume width increases with increasing applied voltage owing to the stronger buoyancy forces generated by the larger temperature gradients, which enhance the lateral spreading of the thermal plume. The coated SMA wire exhibits a consistently larger plume width than the uncoated wire, reaching approximately 4.0 mm at 7 V, compared with about 3.1 mm for the uncoated wire (see Figures S4 and S5 in the supplementary material). The increased plume width indicates enhanced heat transfer from the coated wire to the surrounding air, resulting in improved convective cooling and a faster thermo-mechanical response during cyclic actuation.

Further, the thermo-mechanical response and thermal plume characteristics of the coated and uncoated 150 μm diameter SMA wires are investigated. Fig. 11(a) compares the temperature difference during the heating and cooling stages. The coated SMA wire reaches the peak temperature earlier than that of the uncoated wire due to the lower thermal conductivity of the polymer layer. The coated SMA wire exhibits a steeper initial cooling slope, indicating a higher transient cooling rate immediately after the electrical power is switched off. We have already observed that the heating time decreases for the coated wire. The cooling time also decreases for the coated wire from 4.62 s to 4.16 s. The bead-on-fiber configuration does not behave solely as an insulating barrier; instead, it modifies the overall transient thermal response of the actuator. This enhanced cooling behavior is attributed to the bead-like surface morphology of the polymer coating, which increases the effective heat transfer area, disturbs the thermal boundary layer, and promotes buoyancy-driven natural convection. Consequently, the convective heat transfer coefficient increases, resulting in faster heat removal from the SMA wire. As the temperature difference decreases to approximately 15 °C, the cooling curves of the coated and uncoated wires gradually converge because the buoyancy force decreases with the reduction in the temperature gradient, leading to similar heat transfer characteristics near thermal equilibrium.

Fig. 11(b) illustrates the strain evolution during the thermo-mechanical actuation cycle. Both coated and uncoated SMA wires achieve a maximum recoverable strain of approximately 4%, confirming that the polymer coating does not alter the intrinsic martensite-to-austenite phase transformation or the recoverable deformation of the SMA. However, the coated wire exhibits a shorter dwell period near the fully actuated state owing to its accelerated thermal cycle. The enhanced cooling rate enables the reverse phase transformation to initiate earlier, thereby reducing the overall actuation cycle time while maintaining the same actuation strain.

The variation of the average intensity difference during the heating and cooling cycle is shown in Fig. 11(c). The uncoated SMA wire exhibits a higher peak intensity difference than the coated wire, reflecting the influence of the polymer coating. The peak intensity of the coated wire is achieved earlier than that of the uncoated wire due to a reduction in heat transfer during transient heating caused by the additional conduction resistance offered by the polymer coating. The cooling time is also lower for the coated wire due to enhanced convection caused by the three-dimensional flow generated by the polymer bead on the SMA wire. A supplementary video is provided to understand the difference in the thermal plume evolution of the coated and uncoated SMA wires. The description of the video is provided in Section 3 of the supplementary material. The fluctuations in the thermal plume due to the three-dimensional effect produced by the coating beads are clearly evident in the video. It may be noted that the present study considers a specific polymer and coating configuration. Other polymer materials were not systematically investigated, and the effects of coating thickness and morphology were not independently varied. Furthermore, the long-term durability, adhesion, and environmental stability of the polymer coating under repeated thermal cycling were not systematically evaluated. These aspects require further investigation to assess the suitability of the coated SMA actuator for long-term practical applications.

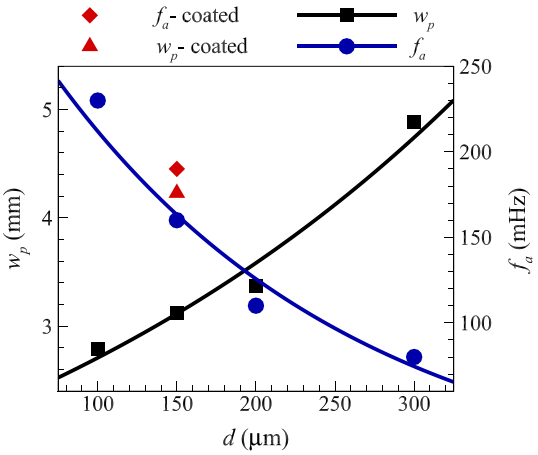


Fig. 12. Comparison of plume width (W_p), and actuation frequency (f_a) as a function of wire diameter (d) for uncoated and coated wires.

3.7. Actuation frequency and efficiency

Fig. 12 shows the comparison of plume width (W_p), and actuation frequency ($f_a = 1/(t_h + t_c)$) as a function of wire diameter (d) for uncoated and coated wires. It indicates that the plume width increases and the actuation frequency decreases with an increase in the diameter of the SMA wire. The larger-diameter SMA wires generate stronger thermal plumes and higher intensity because of greater thermal mass. The smaller-diameter wires have higher actuation frequencies owing to faster heating and cooling response. The improved cooling rate and the corresponding increase in actuation frequency are associated with the coupled thermal and thermo-fluid behavior of the SMA actuator. The heat generated due to electrical actuation raises the SMA temperature above the austenitic transformation temperature and produces the desired shape recovery. After the stoppage of electrical input, the stored thermal energy is dissipated to the surrounding environment before the SMA can undergo the reverse transformation required for the subsequent actuation cycle. The cooling process involves heat conduction within the actuator followed by heat transfer from the external surface of the SMA wire to the surrounding air. The temperature difference between the heated actuator and the surrounding air produces buoyancy-driven flow. The heated air adjacent to the actuator rises and forms a thermal plume, continuously transporting heated fluid away from the actuator and allowing relatively cooler surrounding air to replace it. The strength of this natural-convection process is governed by the temperature difference, the bead geometry, and thermo-physical properties of the surrounding fluid. It may be noted that the cooling stage generally represents a significant portion of the complete thermal cycle and a reduction in cooling time produces a corresponding reduction in the total cycle time i.e., $t_h + t_c$. Since, the actuation frequency is the reciprocal of the total cycle time, any reduction in total cycle time enhances the actuation frequency. The values of t_h, t_c, and f_a are provided in Table 2.

The plume characteristics can have important implications for the design of SMA wire-based actuators employing multiple parallel wires. During Joule heating, the buoyancy-driven thermal plume generated around an SMA wire transports thermal energy into the surrounding fluid. If adjacent wires are positioned too closely, thermal plume interaction may occur, resulting in thermal interference that can potentially influence the heating time, cooling time, phase-transformation behavior, and consequently the actuation frequency of the actuator. Therefore, the spacing between neighboring SMA wires becomes an important design consideration. A detailed comparison of coated and uncoated wires' characteristics is shown in Table 2. The plume width for a 300 μm diameter SMA wire is 4.89 mm, which is substantially larger than its diameter. This indicates that minimum separation between the SMA wires should

Table 2
Comparison of thermal, mechanical, and plume characteristics of SMA wires at an applied voltage of 7 V. Here, d is the SMA wire diameter, t_h is the voltage activation time, t_c is the cooling time, f_a is the actuation frequency, d_s is the displacement, ΔI represents the schlieren intensity, W_p denotes the average plume width, η is the thermo-mechanical energy conversion efficiency, and P_{avg} is the average mechanical output power.

d (μm)	t_h (s)	t_c (s)	f_a (mHz)	d_s (mm)	ΔI (au)	w_p (mm)	η (%)	P_{avg} (mW)
100	1.01	3.36	230	6.02	5.56	2.79	0.18	0.25
150	1.52	4.62	163	5.97	4.72	3.12	0.08	0.18
200	1.86	7.63	106	5.93	4.02	3.37	0.05	0.12
300	2.30	11.03	75	5.92	3.09	4.89	0.02	0.08
150 (Coated)	1.10	4.16	190	5.91	4.30	4.23	0.08	0.21

be governed by the plume width. The comparison between the coated and uncoated 150 μm SMA wires further shows that the polymer-coated wire exhibits a greater plume width and a higher actuation frequency than the uncoated wire, indicating enhanced heat transfer and accelerated thermo-mechanical response. The preliminary design criterion for a multi-wire SMA actuator derived from the measured plume-width data has been presented in the Section 4 of the supplementary material.

The strain energy generated by the SMA actuators is quantified for different wire diameters by calculating the elastic energy stored in the bias spring due to the contraction of the SMA wire during actuation. The thermo-mechanical energy conversion efficiency (η) of the actuators is subsequently determined as the ratio of the stored strain energy (E_{strain}) to the supplied thermal energy ($E_{thermal}$) [46]. The strain energy stored in the bias spring can be calculated as:

$$E_{strain} = \frac{1}{2}\kappa x^2, \tag{8}$$

where κ is the spring stiffness (61.13 N m^{-1}) and x is the measured displacement of the SMA actuator. The supplied thermal energy is given as:

$$E_{thermal} = mc_p\Delta T + \Delta H(\sigma), \tag{9}$$

where m is the mass of the SMA wire, c_p is the specific heat capacity, ΔT is the temperature rise, and $\Delta H(\sigma)$ is the stress-dependent phase transformation enthalpy. The formulation assumes negligible heat exchange with the surroundings and neglects the dependence of the transformation enthalpy on the applied stress. The maximum thermo-mechanical conversion efficiency is observed for the 100 μm diameter SMA wire, i.e., 0.18% compared to 0.02% for 300 μm wire diameter. The average mechanical output power (P_{avg}) is calculated to quantify the rate at which the recoverable mechanical energy is delivered during actuation cycle to the actuator and is expressed as [45,47,48]:

$$P_{avg} = \frac{E_{strain}}{t_h + t_c}, \tag{10}$$

where t_h and t_c denote the voltage activation and cooling times, respectively.

The calculated quantities are summarized in Table 2. A comparison of these values indicates that the thermo-mechanical energy conversion efficiency decreases with increasing wire diameter. The values of average mechanical output power calculated using Eq. (10), indicate that a thin SMA wire delivers mechanical energy at a higher rate. The 100 μm wire exhibited the highest average mechanical output power of 0.25 mW, whereas the 300 μm wire produced only 0.08 mW. Although the coated and uncoated 150 μm SMA wires exhibited nearly identical thermo-mechanical energy conversion efficiencies of approximately 0.08%, the coated wire produced a higher average mechanical output power of 0.21 mW compared to 0.18 mW for the uncoated wire, owing to its shorter actuation cycle, demonstrating an improved dynamic

response without compromising energy conversion efficiency. The identical voltage does not by itself demonstrate identical time-integrated electrical-energy consumption because the instantaneous current and resistance vary with temperature and phase transformation. The present thermo-mechanical efficiency should therefore be interpreted as an indirect estimate of electrical-to-mechanical efficiency or COP.

4. Conclusions

The present study investigates the thermo-mechanical behavior of an SMA actuator. The Schlieren technique is used for visualization of the thermal plume, infrared thermography is used for temperature measurement of the SMA wire, and a displacement sensor is used for strain measurement of the actuator. Polymer coating on the SMA wire is carried out to investigate its effect on the actuator characteristics. The influence of wire diameter on thermal-plume evolution, heat transfer characteristics and actuator performance is systematically investigated using plume width and height evolution, actuation frequency, the heat transfer coefficient, thermo-mechanical energy conversion efficiency and average mechanical power. The principal findings are summarized as follows:

(i) The SMA wire diameter influences the thermo-fluid behavior and dynamic performance of the actuator. The actuation frequency decreases with increasing wire diameter, demonstrating that thinner SMA wires are more suitable for high-frequency actuation. For example, as the wire diameter increases from 100 μm to 300 μm, the cooling time increases from 3.36 s to 11.03 s, while the actuation frequency decreases from 230 mHz to 75 mHz. The heat transfer coefficient decreases with an increase in SMA wire diameter. For example, the heat transfer coefficient of the 100 μm SMA wire is 136% higher than that of the 300 μm SMA wire.

(ii) The maximum thermal plume width increases from 2.79 mm to 4.89 mm with increasing wire diameter from 100 μm to 300 μm. The plume dimensions provide practical guidelines for estimating the minimum inter-wire spacing during the design of multi-wire SMA actuators, thereby minimizing thermal interference between adjacent wires.

(iii) The normalized thermal plume width and normalized plume height show a universal exponential correlation, independent of wire diameter. This correlation provides a generalized framework for characterizing buoyancy-driven thermal plumes generated by SMA wires under natural convection conditions.

(iv) The polymer-coated SMA wire has a higher actuation frequency than the uncoated wire. The polymer coating increases the heating rate due to the additional thermal resistance offered by the polymer coating. Similarly, the coated SMA wire has faster cooling due to the three dimensional effect of the beads. Both coated and uncoated wires exhibit nearly identical thermo-mechanical energy conversion efficiency. However, the coated wire produces a higher average mechanical output power as the same recoverable mechanical energy is delivered within a shorter actuation cycle due to faster heating and cooling rates.

(v) The highest thermo-mechanical energy conversion efficiency of 0.18% is obtained for the 100 μm SMA wire, whereas the 300 μm wire exhibits the lowest efficiency of 0.02%. The average mechanical output power increases from 0.08 mW for the 300 μm wire to 0.25 mW for the 100 μm wire, indicating that thinner SMA wires deliver useful mechanical energy at a substantially higher rate.

Overall, the present study provides a quantitative understanding of the coupling between buoyancy-driven thermal plumes, heat transfer, and the thermo-mechanical performance of the SMA wire actuators.

CRediT authorship contribution statement

Raushan Kumar: Writing – original draft, Visualization, Validation, Methodology, Formal analysis. **Bal Krishan Mishra:** Writing – review & editing, Validation, Resources, Methodology, Formal analysis. **Pradipta Kumar Panigrahi:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis. **Bijaya Bikram Samal:** Writing – review & editing, Visualization, Validation, Methodology, Formal analysis. **Ankur Dwivedi:** Writing – review & editing, Methodology, Formal analysis. **Bishakh Bhattacharya:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships that may be considered potential competing interests:

Bishakh Bhattacharya reports that financial support, administrative support, equipment, drugs, or supplies, and statistical analysis were provided by Portescap. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the financial support received from Portescap India Pvt. Ltd. through the Corporate Social Responsibility Grant (PIPL/DORA/2020022) and from the Central Power Research Institute, Ministry of Power, Government of India under Project No. CPRI/ME/2024003.

Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.sna.2026.118654.

Data availability

Data will be made available on request.

References

[1] E. Makino, T. Mitsuya, T. Shibata, Fabrication of TiNi shape memory micropump, Sens. Actuators A Phys. 88 (3) (2001) 256–262, https://doi.org/10.1016/S0924-4247(00)00522-7

[2] D.J. Bell, T.J. Lu, N.A. Fleck, S.M. Spearing, MEMS actuators and sensors: observations on their performance and selection for purpose, J. Micromech. Microeng. 15 (7) (2005) 153–164, https://doi.org/10.1088/0960-1317/15/7/022

[3] A.P. Jardine, J.D. Bartley-Cho, J.S. Flanagan, Improved design and performance of the SMA torque tube for the DARPA smart wing program, in: Smart Structures and Materials 1999: Industrial and Commercial Applications of Smart Structures Technologies, vol. 3674, 1999, pp. 260–269, https://doi.org/10.1117/12.351564

[4] R. Chi, H. Li, D. Shen, Z. Hou, B. Huang, Enhanced P-Type control: indirect adaptive learning from set-point updates, IEEE Trans. Autom. Control 68 (3) (2023) 1600–1613, https://doi.org/10.1109/TAC.2022.3154347

[5] M. El Dib, R. Gorbet, E. Kubica, X. Gao, A. Browne, N. Johnson, Adaptive SMA actuator priming using resistance feedback, Smart Mater. Struct. 20 (11) (2011) 115018, https://doi.org/10.1088/0964-1726/20/11/115018

[6] J.-H. Guan, Y.-C. Pei, S. Wang, An experimental investigation on time response characteristics of SMA wire actuators under electric heating for engineering design, Smart Mater. Struct. 29 (10) (2020) 105015, https://doi.org/10.1088/1361-665X/aba48e

[7] J. Qiu, J. Tani, D. Osanai, Y. Urushiyama, D. Lewinnek, High-speed response of SMA actuators, Int. J. Appl. Electromagn. Mech. 12 (1–2) (2000) 87–100, https://doi.org/10.3233/JAE-2000-199

[8] I.W. Hunter, S. Lafontaine, J. Hollerbach, P. Hunter, Fast reversible NiTi fibers for use in microrobotics, in: Proc. 1991 IEEE Micro Electro Mechanical Systems—MEMS, vol. 91, 1991, pp. 166–170, https://doi.org/10.1109/MEMSYS.1991.114789

[9] A. Bhattacharyya, L. Sweeney, M. Faulkner, Experimental characterization of free convection during thermal phase transformations in shape memory alloy wires, Smart Mater. Struct. 11 (3) (2002) 411–422, https://doi.org/10.1088/0964-1726/11/3/312

[10] H. Basaeri, A. Yousefi-Koma, M.R. Zakerzadeh, S.S. Mohtasebi, Experimental study of a bio-inspired robotic morphing wing mechanism actuated by shape memory alloy wires, Mechatronics 24 (8) (2014) 1231–1241, https://doi.org/10.1016/j.mechatronics.2014.10.010

[11] J. Jeong, K. Hyeon, J. Han, C.H. Park, S.-Y. Ahn, S.-K. Bok, K.-U. Kyung, Wrist assisting soft wearable robot with stretchable coolant vessel integrated SMA muscle,

IEEE/ASME Trans. Mechatron. 27 (2) (2022) 1046–1058, https://doi.org/10.1109/TMECH.2021.3078472
[12] S.-M. Jeong, D.-H. Kim, D.-W. Oh, Flow visualisation of natural convection around a heated vertical plate using background oriented schlieren, Int. J. Nanotechnol. 22 (1) (2025) 78–86, https://doi.org/10.1504/IJNT.2025.152234
[13] C.H. Park, K.J. Choi, Y.S. Son, Shape memory alloy-based spring bundle actuator controlled by water temperature, IEEE/ASME Trans. Mechatron. 24 (4) (2019) 1798–1807, https://doi.org/10.1109/TMECH.2019.2928881
[14] A. Osorio Salazar, Y. Sugahara, D. Matsuura, Y. Takeda, Scalable output linear actuators, a novel design concept using shape memory alloy wires driven by fluid temperature, Machines 9 (1) (2021) 14, https://doi.org/10.3390/machines9010014
[15] R. Kumar, A. Dwivedi, B. Bhattacharya, Experimental analysis of cooling time reduction in SMA actuator through tuning of wire geometry and heat transfer process, Smart Mater. Struct. 35 (2) (2026) 025009, https://doi.org/10.1088/1361-665X/ae3502
[16] K.L. Chaurasiya, N. Gupta, F. Javeed, V. Kumar, B. Bhattacharya, Enhancing actuation frequency of shape memory alloy-based system with a novel evaporative cooling technique for fast cyclic applications, in: Active and Passive Smart Structures and Integrated Systems XVIII, vol. 12946, 2024, pp. 457–466, https://doi.org/10.1117/12.3006378
[17] S. Oh, R. Tabassian, P. Thangasamy, M. Mahato, V.H. Nguyen, S. Nam, Z. Huapeng, I.-K. Oh, Cooling-accelerated nanowire-nitinol hybrid muscle for versatile prosthetic hand and biomimetic retractable claw, Adv. Funct. Mater. 32 (18) (2022) 2111145, https://doi.org/10.1002/adfm.202111145
[18] R. Kumar, A. Dwivedi, B. Bhattacharya, Optimized cooling time of shape memory alloy wires using GD-assisted magnetocaloric effect, in: Proceedings of the ASME International Mechanical Engineering Congress and Exposition–India (IMECE-India 2025), vol. 89145, American Society of Mechanical Engineers (ASME), India, 2025, pp. 002–02030, https://doi.org/10.1115/IMECE-INDIA2025-160596
[19] K. Akash, A.K. Shukla, S.M. Prabu, D.C. Narayane, S. Kanmanisubbu, I. Palani, Parametric investigations to enhance the thermomechanical properties of CuAlNi shape memory alloy bi-morph, J. Alloys Compd. 720 (2017) 264–271, https://doi.org/10.1016/j.jallcom.2017.05.255
[20] A. Boccaccini, C. Peters, J. Roether, D. Eifler, S. Misra, E. Minay, Electrophoretic deposition of polyetheretherketone (PEEK) and PEEK/Bioglass® coatings on NiTi shape memory alloy wires, J. Mater. Sci. 41 (24) (2006) 8152–8159, https://doi.org/10.1007/s10853-006-0556-z
[21] F. Marashi-Najafi, J. Khalil-Allafi, S. Mahdavi, Superior multifunctional polypyrrole anticorrosion coating modified by polydopamine decorated barium titanate nanoparticles on NiTi shape memory alloys, J. Mater. Res. Technol. 26 (2023) 6823–6841, https://doi.org/10.1016/j.jmrt.2023.09.071
[22] S. Samal, O. Kosjakova, D. Vokoun, I. Stachiv, Shape memory behaviour of PMMA-coated NiTi alloy under thermal cycle, Polymers 14 (14) (2022) 2932, https://doi.org/10.3390/polym14142932
[23] M. Perez-Zúñiga, F. Sánchez-Arévalo, J. Hernández-Cordero, Enhanced photomechanical response of a Ni–Ti shape memory alloy coated with polymer-based photothermal composites, Smart Mater. Struct. 26 (10) (2017) 105012, https://doi.org/10.1088/1361-665X/aa80cb
[24] C.T. Gabbard, J.B. Bostwick, Bead-on-fibre morphology in shear-thinning flow, J. Fluid Mech. 961 (2023) 14, https://doi.org/10.1017/jfm.2023.248
[25] C.W. Rodricks, I. Greenfeld, B. Fiedler, H.D. Wagner, Fragmentation of beaded fibres in a composite, Materials 15 (3) (2022) 890, https://doi.org/10.3390/ma15030890
[26] A. Pour Karimi, B. Hormann, H. Askarizadeh, W. Rohlfs, B. Scheid, R. Kneer, Drop-off characteristics of film flow on inclined fibers, Eur. Phys. J. Spec. Top. 234 (11) (2025) 3153–3178, https://doi.org/10.1140/epjs/s11734-024-01457-z
[27] C.T. Gabbard, J.T. Rhoads, J.-M. Delhaye, J.B. Bostwick, Scaling analysis for the frequency of ostwaldian and newtonian bead-on-fibre flows, J. Fluid Mech. 1002 (2025) 12, https://doi.org/10.1017/jfm.2024.1176
[28] R. Britz, P. Motzki, Analysis and evaluation of bundled SMA actuator wires, Sens. Actuators A Phys. 333 (2022) 113233, https://doi.org/10.1016/j.sna.2021.113233
[29] R. Barakat, S.-M. Kirsch, F. Welsch, P. Motzki, Bidirectional rotational antagonistic shape memory alloy actuators for high-frequency artificial muscles, Sci. Rep. 15 (1) (2025) 9108, https://doi.org/10.1038/s41598-025-93209-9
[30] K.L. Chaurasiya, A.S. Harsha, Y. Sinha, B. Bhattacharya, Design and development of non-magnetic hierarchical actuator powered by shape memory alloy based bipennate muscle, Sci. Rep. 12 (1) (2022) 10758, https://doi.org/10.1038/s41598-022-14848-w
[31] M. Fossa, M. Misale, G. Tanda, Schlieren visualization of natural convection flows, J. Phys. Conf. Ser. 655 (2015) 012005, https://doi.org/10.1088/1742-6596/655/1/012005
[32] B. Gebhart, Y. Jaluria, R.L. Mahajan, B. Sammakia, Buoyancy-Induced Flows and Transport, Hemisphere Publishing Corporation, New York, NY, 1988.
[33] M. Raffel, Background-oriented schlieren (BOS) techniques, Exp. Fluids 56 (3) (2015) 60, https://doi.org/10.1007/s00348-015-1927-5
[34] L. Venkatakrishnan, G.E.A. Meier, Density measurements using the background oriented schlieren technique, Exp. Fluids 37 (2) (2004) 237–247, https://doi.org/10.1007/s00348-004-0807-1
[35] J. Mohd, D. Das, Volumetric density measurement in buoyant plumes using tomographic background-oriented schlieren (TBOS), arXiv preprint arXiv:2503.17705, 2025.
[36] C. Yang, X. Xu, J. Xiong, L. Chen, J. Ma, A. Li, Flow and evolution characteristics of the thermal plumes above an inclined area heat source, Energy Built Environ. (2025), https://doi.org/10.1016/j.enbenv.2025.05.010
[37] J. Ramaiah, T. Rubeis, R. Gannavarpu, D. Ambrosini, Quantitative flow visualization by hidden grid background oriented schlieren, Opt. Lasers Eng. 160 (2023) 107307, https://doi.org/10.1016/j.optlaseng.2022.107307
[38] I. Mellouki, N. Bennaji, N. Yacoubi, IR characterization of graphite black-coating for cryogenic detectors, Infrared Phys. Technol. 50 (1) (2007) 58–62, https://doi.org/10.1016/j.infrared.2006.05.002
[39] S. Shu, T. Wu, Z. Wang, Y. Zhang, Z. Yang, H. Liang, Research on the normal emissivity of graphite between 150 and 500° C by an infrared camera for nuclear fusion devices, Nucl. Mater. Energy 31 (2022) 101182, https://doi.org/10.1016/j.nme.2022.101182
[40] Q.Y. Hamid, W.Z. Wan Hasan, M.A. Azmah Hanim, A.A. Nuraini, M.N. Hamidon, Multi-sensor thermo–electro–mechanical characterization of an SMA wire for wearable actuation design, J. Eng. Res. (2026), https://doi.org/10.1016/j.jer.2026.07.013
[41] D. Quéré, Fluid coating on a fiber, Annu. Rev. Fluid Mech. 31 (1) (1999) 347–384, https://doi.org/10.1146/annurev.fluid.31.1.347
[42] J. Eggers, E. Villermaux, Physics of liquid jets, Rep. Prog. Phys. 71 (3) (2008) 036601, https://doi.org/10.1088/0034-4885/71/3/036601
[43] S.K. Singh, P.K. Panigrahi, K. Muralidhar, Effect of buoyancy on the wakes of circular and square cylinders: a schlieren-interferometric study, Exp. Fluids 43 (1) (2007) 101–123, https://doi.org/10.1007/s00348-007-0329-8
[44] S.W. Churchill, H.H.S. Chu, Correlating equations for laminar and turbulent free convection from a horizontal cylinder, Int. J. Heat Mass Transf. 18 (9) (1975) 1049–1053, https://doi.org/10.1016/0017-9310(75)90222-7
[45] P.L. Potapov, E.P. Da Silva, Time response of shape memory alloy actuators, J. Intell. Mater. Syst. Struct. 11 (2) (2000) 125–134, https://doi.org/10.1106/XH1H-FH3Q-1YEX-4H3F
[46] P.C.C. Monteiro Jr, L.L. Silva, T.A. Netto, M.A. Savi, Experimental investigation of the influence of the heating rate in an SMA actuator performance, Sens. Actuators A Phys. 199 (2013) 254–259, https://doi.org/10.1016/j.sna.2013.05.016
[47] A. Dana, S. Vollach, D. Shilo, Use the force: review of high-rate actuation of shape memory alloys, Actuators 10 (7) (2021), https://doi.org/10.3390/act10070140
[48] D. Avirovik, R.A. Kishore, D. Vuckovic, S. Priya, Miniature shape memory alloy heat engine for powering wireless sensor nodes, Energy Harvest. Syst. 1 (1–2) (2014) 13–18, https://doi.org/10.1515/ehs-2013-0003